

# Machine Learning Assignment 1

## Polynomial Regression Report

### Analysis of Phase I & II

Name: Sushmit Biswas | Roll Number: BT2024038

**Abstract:** This report documents the process of building Polynomial Regression models for two personalized datasets. The methodology uses exploratory data analysis (EDA) for descriptive correlation analysis, followed by five-fold cross-validation to select the polynomial degree. By evaluating Out-of-Fold (OOF) predictions, the selected models achieved out-of-fold  $R^2$  estimates above 0.90.

## 1 Analytical Premise

Standard polynomial approximations rely on  $w_0 + \sum_{i=1}^d w_i \phi_i(\mathbf{x})$ . Selecting the polynomial degree without validation can cause underfitting at low degrees and severe overfitting or numerical instability at high degrees. To reduce the risk of overfitting, polynomial degrees were selected using five-fold shuffled cross-validation with `random_state = 42`, based on validation mean squared error. All available input features were retained, allowing the polynomial expansion to represent nonlinear and interaction effects.

## 2 Phase 1: Steam Turbine Optimization (var1)

The goal of Phase 1 is estimating the Net Power Score ( $y$ ) based on physical turbine variables.

### 2.1 Inclusive Feature Representation

A Pearson correlation matrix was computed for the available features and target for exploratory analysis. No features were removed based on marginal correlation.

### 2.2 Cross-Validation

Using all six available features, five-fold cross-validation evaluated candidate degrees from 1 through 10. Degree 3 produced the lowest observed validation MSE among the candidate degrees 1–10. At higher degrees, training MSE decreased sharply while validation MSE increased, indicating overfitting. From degree 6 onward, the number of polynomial terms becomes comparable to or larger than the number of observations, causing near-zero training error while validation error remains substantially higher. Therefore, validation MSE rather than training MSE was used for degree selection.

### 2.3 Phase 1 Empirical Results

- **Features Used:**  $x_1, x_2, x_3, x_4, x_5, x_6$
- **Selected Degree:** 3
- **Training MSE:** 0.7727, **Training  $R^2$ :** 0.9272
- **Out-of-fold MSE:** 0.9895
- **Out-of-fold  $R^2$ :** 0.9067

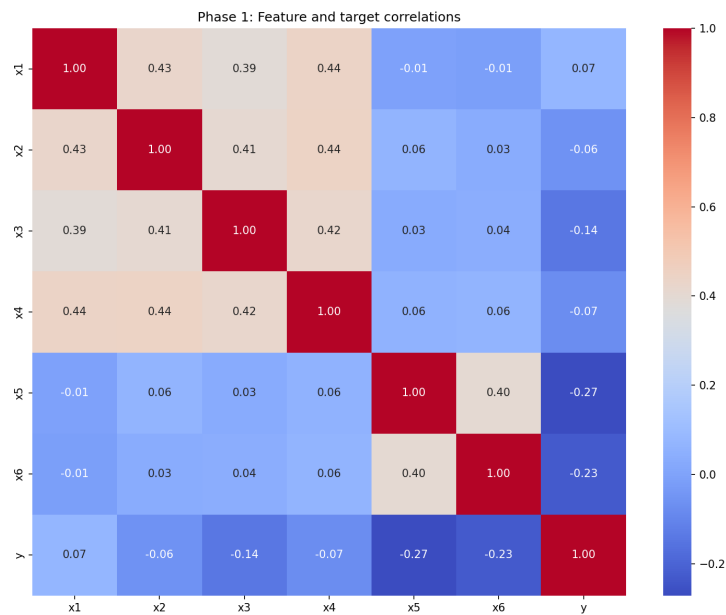


Figure 1: Pearson Correlation Heatmap displaying relational bounds in Phase 1.

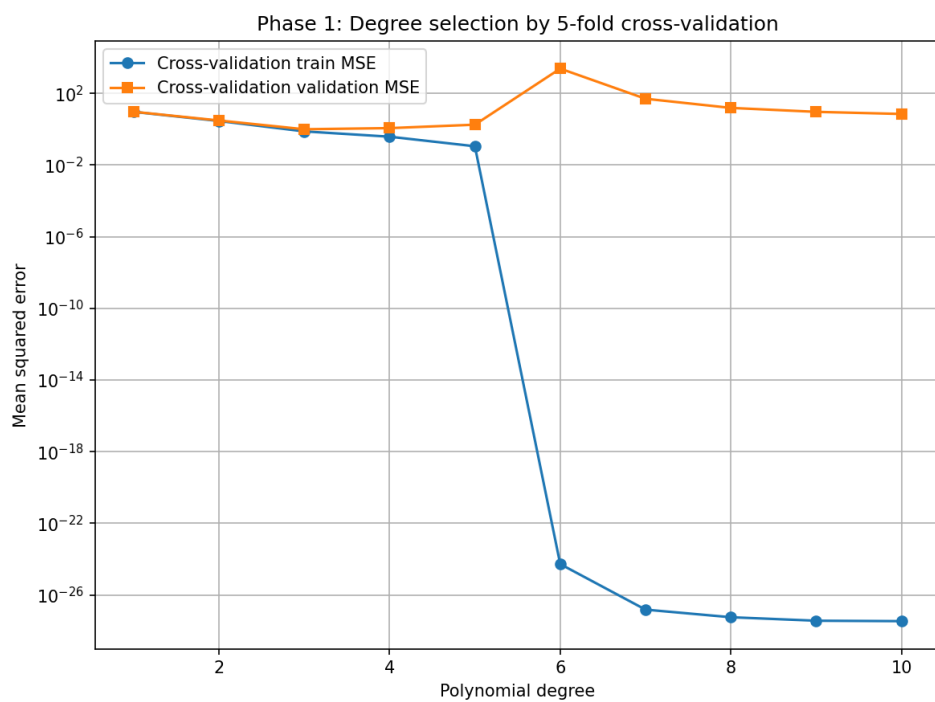


Figure 2: Five-fold cross-validation MSE across candidate polynomial degrees.

These values indicate that the selected polynomial captures a substantial portion of the observed target variation. Residuals are broadly centered around zero, with some outliers and no dominant visible structure.

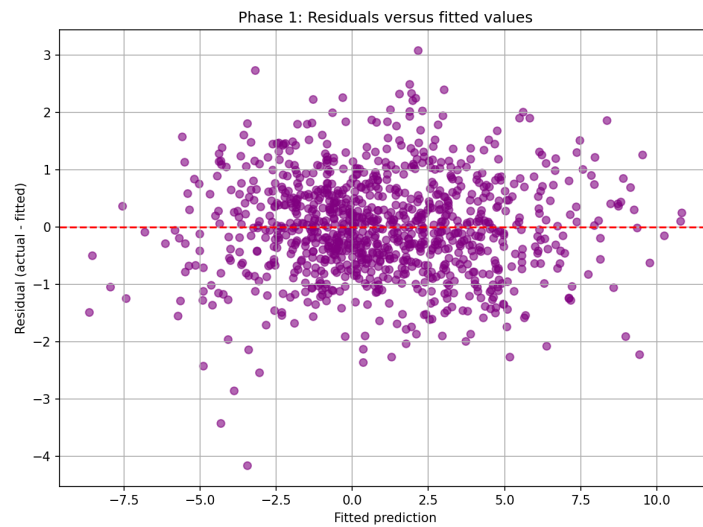


Figure 3: Residual distribution mapping relative predictive error across constraints.

### 3 Phase 2: Subterranean Thermal Mapping (var2)

#### 3.1 Automated Structural Mapping

The spatial coordinates may exhibit nonlinear and interaction effects that are not captured by marginal correlations alone. Again, all three available coordinates were retained so that the

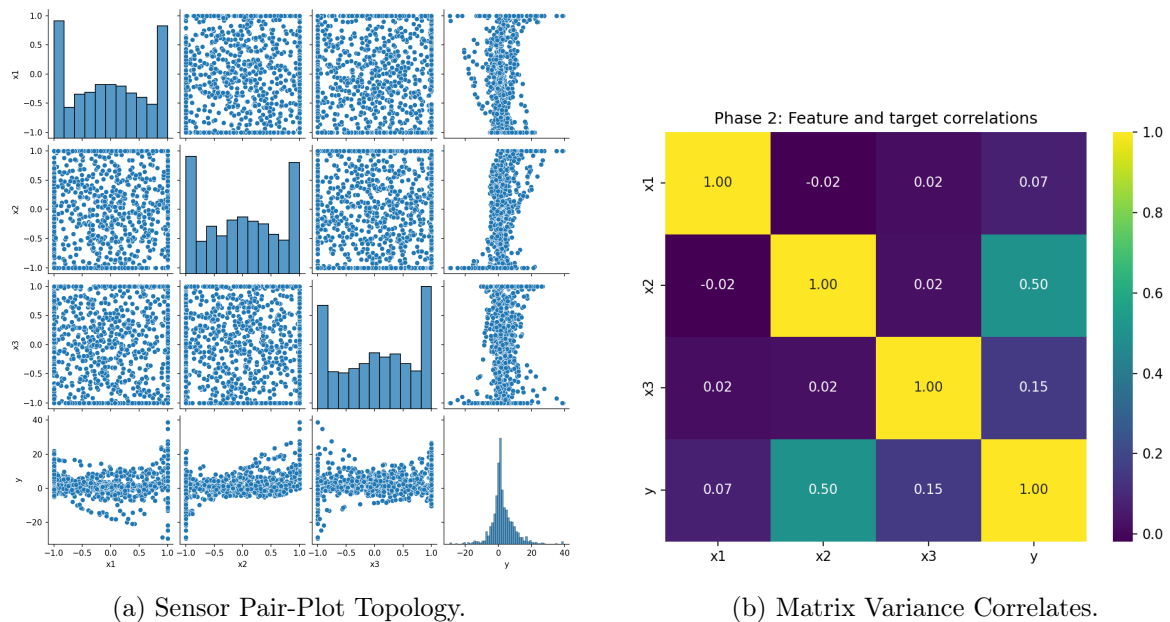


Figure 4: Feature evaluations confirming mathematical retention bounds.

polynomial model could represent their nonlinear and interaction effects.

### 3.2 Choosing the Polynomial Degree

Candidate degrees from 1 through 20 were evaluated using five-fold cross-validation. The plotted MSE values use a logarithmic y-axis because the validation error spans several orders of magnitude. Degree 8 produced the lowest observed validation MSE among the candidate degrees 1–20 and

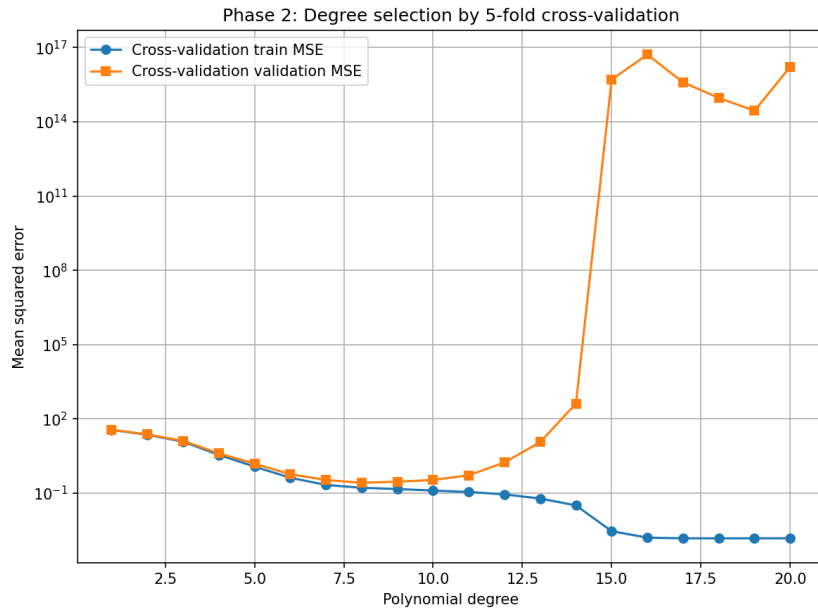


Figure 5: Five-fold cross-validation MSE across candidate polynomial degrees.

was therefore selected.

### 3.3 Phase 2 Empirical Results

- **Features Used:**  $x_1, x_2, x_3$
- **Selected Degree:** 8
- **Training MSE:** 0.1731, **Training  $R^2$ :** 0.9965
- **Out-of-fold MSE:** 0.2629
- **Out-of-fold  $R^2$ :** 0.9947

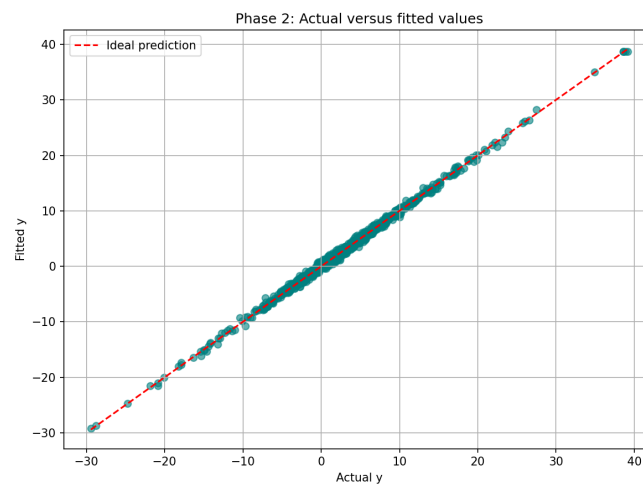


Figure 6: True output mappings accurately predicting geological boundaries.

## 4 Overfitting at High Degrees

The degree searches evaluated the specified candidate ranges: degrees 1–10 for Phase 1 and degrees 1–20 for Phase 2. These experiments also illustrate the computational and statistical risks of high-degree polynomial expansions. At high degrees, the number of polynomial terms can become comparable to or larger than the number of observations, resulting in severe overparameterization. For Phase 1, degree 10 with six input features produces 8,007 nonconstant polynomial terms. With approximately 1,000 training observations, the expanded design matrix can become underdetermined and produce nearly zero training error.

However, a high-dimensional polynomial surface can produce unstable and inaccurate predictions on held-out data, consistent with the large validation-error spikes observed in the right tails of the CV curves. Therefore, validation MSE was used to select degree 3 for Phase 1 and degree 8 for Phase 2. No Ridge or Lasso regularization was used in the final models.

## 5 Conclusion

The results show that retaining all available features and selecting the polynomial degree using five-fold cross-validation produced strong validation estimates. Phase 1 selected degree 3 and achieved an out-of-fold  $R^2$  of 0.9067, while Phase 2 selected degree 8 and achieved an out-of-fold  $R^2$  of 0.9947. These values are cross-validation estimates and should not be interpreted as hidden-test performance.

**Code Availability:** The complete source code and reproducible pipelines for this assignment can be found at <https://github.com/Sushmit-Biswas/Polynomial-Regression-ML-Assignment>.